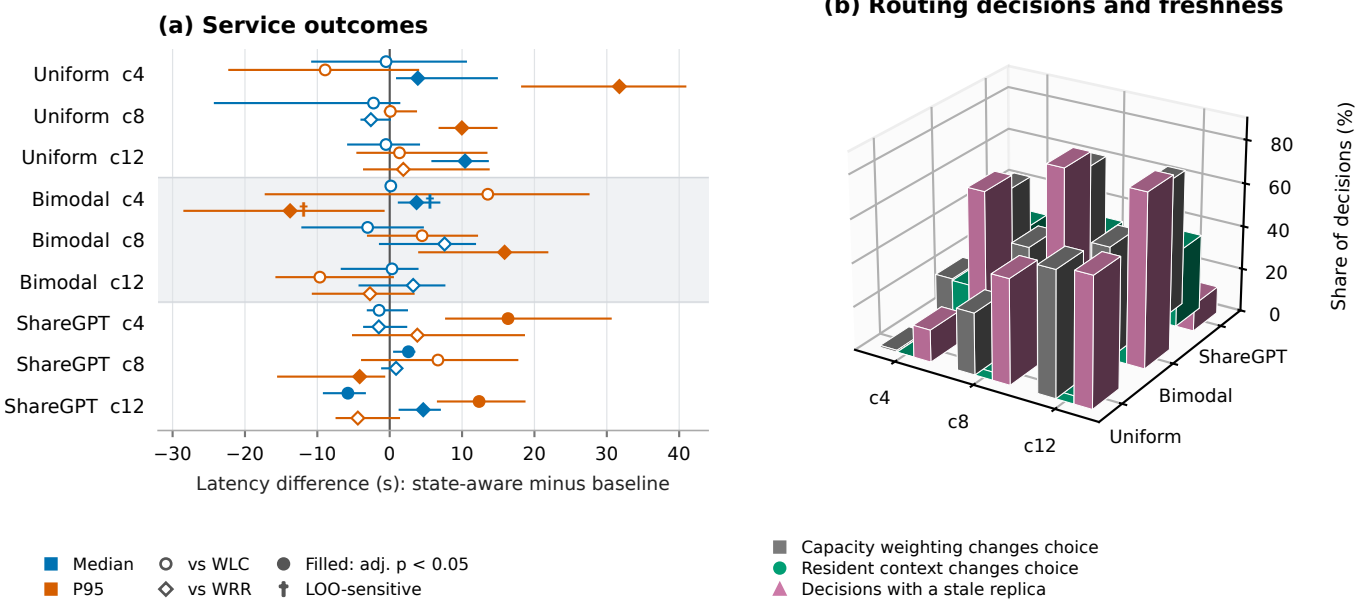
(a) Service outcomes
Uniform c4
Uniform c8
Uniform c12
Bimodal c4
Bimodal c8
Bimodal c12
ShareGPT c4
ShareGPT c8
ShareGPT c12
−30
−20
−10
0
10
20
30
40
Latency difference (s): state-aware minus baseline
Median
P95
vs WLC
vs WRR
Filled: adj. p < 0.05
LOO-sensitive
(b) Routing decisions and freshness
Share of decisions (%)
0
20
40
60
80
c4
c8
c12
Uniform
Bimodal
ShareGPT
Capacity weighting changes choice
Resident context changes choice
Decisions with a stale replica